

#6 Aplique el método de Gauss - Jordan para determinar el conjunto solución del sistema: (4 Pts)

$$\begin{cases} x + y + z + w = 5 \\ 2x + 3y + 2z + 2w = 13 \\ 3x + 4y + 3z + 3w = 18 \end{cases}$$

$$\left( \begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 2 & 3 & 2 & 2 & 13 \\ 3 & 4 & 3 & 3 & 18 \end{array} \right)$$

$$\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}$$

$$\begin{array}{l} -2 \cdot F_1 + \widetilde{F_2} \\ -3 \cdot F_1 + \widetilde{F_3} \end{array} \left( \begin{array}{cccc|c} 1 & 1 & 1 & 1 & 5 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 & 3 \end{array} \right)$$

$$\begin{array}{l} -F_2 + \widetilde{F_1} \\ -F_2 + \widetilde{F_3} \end{array} \left( \begin{array}{cccc|c} x & y & z & w & s \\ 1 & 0 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$x + z + w = 2 \rightarrow x = 2 - z - w$$

$$y = 3$$

$$S = \{ (2 - z - w, 3, z, w) \}$$

5. Considere las matrices  $A$  y  $X$  de dimensiones  $3 \times 3$  y la matriz  $B$  de dimensión  $2 \times 3$ , para las cuales está definida la igualdad

$$XA + B^T B = 2X.$$

$2I - A$  es invertible

- a) Utilice álgebra matricial para despejar  $X$  de la ecuación anterior. (3 puntos),

$$XA + B^T B = 2X$$

$$AI = A$$

$$B^T B = 2X - \underline{XA}$$

$$2X - AX$$

$$(2I - A)X$$

$$B^T B = X \cdot 2 - XA$$

$$K \cdot A = A \cdot K$$

$$B^T B = X(2I - A)$$

$$AX = B$$

$$X = B \cdot A^{-1}$$

$$\boxed{X = (B^T B) \cdot (2I - A)^{-1}}$$

- b) Determine  $X$  de forma explícita, si  $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}$  y  $B = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{pmatrix}$  (4 puntos)

$$B^T = \begin{pmatrix} 1 & -1 \\ 0 & 2 \\ 1 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 1 \\ -1 & 2 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 1+1 & 0-2 & 1+2 \\ 0-2 & 0+4 & 0-4 \\ 1+2 & 0-4 & 1+4 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -2 & 3 \\ -2 & 4 & -4 \\ 3 & -4 & 5 \end{pmatrix}$$

$$(2I-A), A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}, I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$2I = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} - A$$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 & -1 \\ 0 & -2 & -3 \\ -1 & 1 & 2 \end{pmatrix}$$

$$2I-A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 4 & 3 \\ 1 & -1 & 0 \end{pmatrix}$$

$$(2I-A)^{-1} = \left( \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 4 & 3 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 \end{array} \right)$$

$$F_2 \leftrightarrow F_3 \left( \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$-F_1 + \widetilde{F}_2 \left( \begin{array}{ccc|ccc} \boxed{1} & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$\left( \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 & 0 & 1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$-1 \cdot \widetilde{F}_2 \left( \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 4 & 3 & 0 & 1 & 0 \end{array} \right)$$

$$-4 \cdot \widetilde{F}_2 + \widetilde{F}_3 \left( \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 0 & -1 & -4 & 1 & 4 \end{array} \right)$$

$$-1 \cdot \widetilde{F}_3 \left( \begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 0 & 1 & 4 & -1 & -4 \end{array} \right)$$

$$\begin{array}{l} -F_3 + \widetilde{F}_1 \\ -F_3 + \widetilde{F}_2 \end{array} \left( \begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & 4 \\ 0 & 1 & 0 & -3 & 1 & 3 \\ 0 & 0 & 1 & 4 & -1 & -4 \end{array} \right)$$

$$(2I - A)^{-1} = \begin{pmatrix} -3 & 1 & 4 \\ -3 & 1 & 3 \\ 4 & -1 & -4 \end{pmatrix}$$

$$(B^T B) \cdot (2I - A)^{-1}$$

$$\begin{pmatrix} 2 & -2 & 3 \\ -2 & 4 & -4 \\ 3 & -4 & 5 \end{pmatrix} \times \begin{pmatrix} -3 & 1 & 4 \\ -3 & 1 & 3 \\ 4 & -1 & -4 \end{pmatrix}$$

$$\begin{pmatrix} \cancel{-6} + \cancel{6} + 12 & \cancel{2} - \cancel{2} - 3 & 0 - 6 - 12 \\ 6 - 12 - 16 & -2 + 4 + 4 & -8 + 12 + 16 \\ -9 + 12 + 20 & 3 - 4 - 5 & \cancel{12} - \cancel{12} - 20 \end{pmatrix}$$

$$\begin{pmatrix} 12 & -3 & -12 \\ -22 & 6 & 20 \\ 23 & -6 & -20 \end{pmatrix}$$

6. Determine el número complejo  $z$  que satisface simultáneamente las condiciones siguientes: (4 puntos)

$$z = a + bi \quad \begin{cases} |z-1| = 2 \\ \text{Arg}(z-2-i) = -\frac{\pi}{2} \end{cases} \rightarrow \pm \frac{\pi}{2} \quad \begin{matrix} a=0 \\ b < 0 \end{matrix}$$

$$a + bi - 2 - i \quad \pm \pi, \frac{\pi}{2}$$

$$\begin{matrix} (a-2) + (b-1)i \\ a & b \\ a=0 & b < 0 \end{matrix} \quad \frac{b}{a} = \tan(\theta)$$

$$\begin{matrix} a-2=0 & b-1 < 0 \\ a=2 & b < 1 \end{matrix}$$

$$|z-1| = 2$$

$$|a + bi - 1| = 2$$

$$|(a-1) + (b)i| = 2$$

$$(a-1)^2 + b^2 = 2^2 \quad a=2$$

$$(2-1)^2 + b^2 = 4$$

$$1 + b^2 = 4$$

$$b^2 = 3$$

$$b = \pm \sqrt{3}$$

$$b = -\sqrt{3}$$

$$b < 0$$

$$\boxed{R1 \quad 2 - \sqrt{3}i}$$

Rectangular

8. Expresa  $(-1 - i)^i$  en la forma  $a + bi$ , con  $a$  y  $b$  números reales.

$$z = (-1 - i)^i$$

$$\ln(x)^h = h \ln(x)$$

$$\ln(z) = \ln(-1 - i)^i$$

$$r e^{\theta i}$$

$$i \ln(-1 - i)$$

$$r = \sqrt{a^2 + b^2}$$

$$\theta = \arctan\left(\frac{b}{a}\right) \pm \pi$$

$$z = \begin{matrix} a & b \\ -1 & -1 \end{matrix}$$

$$r = \sqrt{(-1)^2 + (-1)^2} = \sqrt{2}$$

$$a < 0 \quad b > 0 \rightarrow +\pi$$

$$a < 0 \quad b < 0 \rightarrow -\pi$$

$$\theta = \arctan\left(\frac{-1}{-1}\right) - \pi = -\frac{3\pi}{4}$$

$$a = 0 \quad b > 0 \rightarrow \theta = \frac{\pi}{2}$$

$$a = 0 \quad b < 0 \rightarrow \theta = -\frac{\pi}{2}$$

$$\sqrt{2} \cdot e^{-\frac{3\pi}{4} i}$$

$$i \ln(\sqrt{2} \cdot e^{-\frac{3\pi}{4} i})$$

$$r e^{\theta i}$$

$$i \ln(\sqrt{2}) + i \ln(e) - \frac{3\pi}{4} i$$

$$\ln(a \cdot b) = \ln(a) + \ln(b)$$

$$\ln(\sqrt{2}) i + -\frac{3\pi}{4} i$$

$$\ln(e)^x = x$$

$$\ln(\sqrt{2}) i + \frac{3\pi}{4}$$

$$e^{\ln(\sqrt{2}) i} + e^{\frac{3\pi}{4}}$$

$$r e^{\theta i} \rightarrow r \operatorname{cis}(\theta)$$

$$\operatorname{cis}(\ln(\sqrt{2})) + e^{\frac{3\pi}{4}}$$

$$\text{cis} = \cos + i \sin$$

$$\text{cis}(\ln(\sqrt{2})) + e^{\frac{3\pi}{4}}$$

$$[\cos(\ln(\sqrt{2})) + i \sin(\ln(\sqrt{2}))] \cdot e^{\frac{3\pi}{4}}$$

$$\cos(\ln(\sqrt{2})) \cdot e^{\frac{3\pi}{4}} + \sin(\ln(\sqrt{2})) \cdot e^{\frac{3\pi}{4}} i$$

$$\boxed{9.92 + 3.58 i}$$



$$\frac{(1-i)^6 (1-\sqrt{3}i)^3}{(-2\sqrt{3}+2i)^7}$$

$$r = \sqrt{a^2 + b^2}$$

$$\theta = \arctan\left(\frac{b}{a}\right) \begin{matrix} +\pi \\ -0 \end{matrix}$$

$$r_1 \cdot \text{cis}(\theta_1) \cdot r_2 \cdot \text{cis}(\theta_2)$$

$$a < 0 \quad b > 0 \rightarrow +\pi$$

$$a < 0 \quad b < 0 \rightarrow -\pi$$

$$r_1 \cdot r_2 \cdot \text{cis}(\theta_1 + \theta_2)$$

$$a = 0 \quad b > 0 \rightarrow \theta = \frac{\pi}{2}$$

$$a = 0 \quad b < 0 \rightarrow \theta = -\frac{\pi}{2}$$

$$\frac{r_1 \cdot \text{cis}(\theta_1)}{r_2 \cdot \text{cis}(\theta_2)} = \frac{r_1}{r_2} \cdot \text{cis}(\theta_1 - \theta_2)$$

$$(r \text{cis}(\theta))^n = r^n \cdot \text{cis}(\theta \cdot n)$$

$$\frac{(1-i)^6 (1-\sqrt{3}i)^3}{(-2\sqrt{3}+2i)^7}$$

$z_1$        $z_2$   
 $z_3$

$$z_1 = 1-i \quad a > 0 \quad b < 0$$

$$r = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

$$\theta = \arctan\left(\frac{-1}{1}\right) = -\frac{\pi}{4}$$

$$z_1 = \sqrt{2} \text{cis}\left(-\frac{\pi}{4}\right)$$

$$z_2 = 1-\sqrt{3}i \quad a > 0 \quad b < 0$$

$$r = \sqrt{1^2 + (-\sqrt{3})^2} = 2$$

$$\theta = \tan^{-1}\left(\frac{-\sqrt{3}}{1}\right) = -\frac{\pi}{3}$$

$$z_2 = 2 \cdot \text{cis}\left(-\frac{\pi}{3}\right)$$

$$z_3 = -2\sqrt{3} + 2i \quad a < 0 \quad b > 0 \\ \rightarrow +\pi$$

$$r = \sqrt{(-2\sqrt{3})^2 + 2^2} = 4$$

$$\theta = \arctan\left(\frac{-2\sqrt{3}}{2}\right) + \pi = \frac{5\pi}{6}$$

$$z_3 = 4 \cdot \text{cis}\left(\frac{5\pi}{6}\right)$$

$$\frac{\overbrace{(1-i)}^{z_1} \overbrace{(1-\sqrt{3}i)}^{z_2}}{\underbrace{(-2\sqrt{3}+2i)}_{z_3}}$$

$$\frac{(\sqrt{2} \cdot \text{cis}(\frac{-\pi}{4}))^6 \cdot (2 \cdot \text{cis}(\frac{-\pi}{3}))^3}{(4 \cdot \text{cis}(\frac{5\pi}{6}))^7}$$

$$\frac{(\sqrt{2})^6 \cdot \text{cis}(\frac{-\pi}{4} \cdot 6) \cdot 2^3 \cdot \text{cis}(\frac{-\pi}{3} \cdot 3)}{4^7 \cdot \text{cis}(\frac{5\pi}{6} \cdot 7)}$$

$$r_1 \cdot \text{cis}(\theta_1) \cdot r_2 \cdot \text{cis}(\theta_2) \\ r_1 \cdot r_2 \cdot \text{cis}(\theta_1 + \theta_2)$$

$$\frac{(\sqrt{2})^6 \cdot 2^3 \cdot \text{cis}(\frac{6\pi}{4} - \pi)}{4^7 \cdot \text{cis}(\frac{-35\pi}{6})}$$

$$\frac{2^3 \cdot 2^3 \cdot \text{cis}(\frac{5\pi}{2})}{4^7 \cdot \text{cis}(\frac{-35\pi}{6})}$$

$$\frac{2^3 \cdot 2^3}{4^7} \cdot \text{cis}\left(\frac{-5\pi}{2} - \frac{-35\pi}{6}\right)$$

$$\frac{1}{2^8} \cdot \text{cis}\left(\frac{-25\pi}{3}\right)$$

$$\frac{I}{2^8} \cdot \cos\left(\frac{-25\pi}{3}\right)$$

$$\frac{I}{2^8} \cdot \left[ \cos\left(\frac{-25\pi}{3}\right) + \sin\left(\frac{-25\pi}{3}\right)i \right]$$

$$\boxed{\frac{I}{2^8} \cdot \cos\left(\frac{-25\pi}{3}\right) + \sin\left(\frac{-25\pi}{3}\right)i - \frac{I}{2^8} \cdot}$$